Problem : Loan Approval System AI agent

AI Agent Hackathon Team Design Sheet

Team Name

AI Musketeers

Team Members

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Project Name

Loan Approval System AI agent

1. Define Goal

Problem Type

Choose one:

- [ ] Knowledge: answer from documents
- [ ] Action: use tools or APIs
- [x] Workflow: manage steps and approvals
- [x] Decision: classify, score, or recommend
- [ ] Productivity: summarize, draft, organize

Selected: Decision: classify, score, or recommend

Agent Mission

What will your agent do?

Selected mission: Evaluate incoming loan applications by analyzing the applicant's credit score, income, and requested loan amount to recommend whether the loan should be approved, rejected, or flagged for manual review.

Target User

Choose one:

- [ ] Developer
- [ ] Business user
- [ ] Support team
- [ ] Sales team
- [x] Internal operations team
- [ ] Other: [Fill here]

Selected: Internal operations team (Loan Officers / Underwriters)

Input

What does the user give the agent?

What does the user give the agent? Natural language text containing the applicant's details, specifically: requested loan amount, annual income, credit score, and employment status.

Output

What should the agent produce?

What should the agent produce? A clear decision recommendation (APPROVED, REJECTED, INCOMPLETE DATA, or MANUAL REVIEW) along with a brief explanation detailing the reasoning based on the calculated credit status and risk level.

Success Metric

Choose 2-3:

- [x] Correct answer
- [ ] Right tool selected
- [x] Task completed
- [x] Human approval captured
- [ ] Time saved
- [ ] Cited evidence
- [ ] Clean final report

Selected:
Task completed (successfully parsing data and reaching a decision)
Correct answer (accurately calculating risk and applying approval logic)

Human approval captured (correctly identifying edge cases that require a manual underwriter review)

Autonomy Level

Choose one:

- [ ] Copilot: suggests only, human acts
- [x] Approval agent: prepares action, waits for approval
- [ ] Semi-autonomous: acts within safe limits
- [ ] Autonomous: acts end-to-end with monitoring

Selected: Approval agent: prepares action, waits for approval (The agent acts as an advanced underwriter assistant that prepares the recommendation, leaving the final legal sign-off to a human).

2. Choose Stack

Model

Choose one or more:

- [x] GPT-5.5 / GPT-5.4
- [ ] Claude Sonnet 4.6
- [ ] Claude Opus 4.7
- [ ] Gemini 3 Pro
- [ ] gpt-oss / open-weight model
- [ ] Other: [Fill here]

Selected: [Fill here]

Why this model? It is extremely fast, highly cost-effective, and exceptionally reliable at extracting structured JSON data (applicant metrics) from natural language, which is the primary LLM workload in this deterministic workflow.

Programming Language

Choose one:

- [x] Python
- [ ] TypeScript
- [ ] Java

- [ ] .NET
- [ ] Other: [Fill here]

Selected: Python

Why? Python offers the most mature ecosystem for AI development, native compatibility with LangGraph, and excellent data processing capabilities.

Agent Framework

Choose one:

- [ ] LangChain create_agent
- [x] LangGraph
- [ ] OpenAI Agents SDK
- [ ] Deep Agents
- [ ] Custom tool-calling loop
- [ ] Other: [Fill here]

Selected: [Fill here]

Why? LangGraph allows us to define the approval process as a strict, cyclic state machine (Graph). This ensures predictable routing where the LLM handles data extraction, but strict deterministic Python logic handles the sensitive financial math (Credit Check & Risk Assessment), eliminating the risk of LLM hallucinations during decision-making.

Interface

Choose one:

- [x] Chatbot
- [ ] Web app
- [ ] CLI
- [ ] API
- [ ] Slack / Teams bot
- [ ] Voice agent
- [ ] Browser agent
- [ ] Background job

Selected:

App Framework

Choose one:

- [ ] Streamlit
- [ ] Gradio
- [ ] FastAPI
- [x] Next.js (frontend + backend with TypeScript) / **React (frontend) + FastAPI (Python backend)**
- [ ] Typer / Click (CLI)
- [ ] Other: [Fill here]

Selected: Gradio allows for rapid prototyping of Chatbot interfaces, natively supports message histories, and matches the tech stack we previously used for your Process Excellence Copilot.

SQLLite (for the MVP) SQLLite

PostgreSQL / MySQL

Tools used : API Calls (Cibil Score checker)

3. Provide Tools And Knowledge

Tools Needed

Select all that apply:

- [ ] Search
- [ ] Calculator
- [x] File / PDF reader
- [x] API call (Credit score API)
- [ ] Browser automation
- [ ] Database
- [ ] Email
- [ ] Calendar
- [ ] CRM
- [ ] Ticketing system

Selected tools: [Fill here]

Knowledge Source

What knowledge will the agent use?

- [x] Uploaded documents
- [ ] Company SOPs
- [ ] Class notes
- [ ] Website pages
- [x] Database
- [ ] No external knowledge

Selected: [Fill here]

RAG Needed?

- [x] Yes
- [ ] No

If yes, what documents? Loan document policy

Vector Store

Choose one if using RAG:

- [ ] Chroma
- [x] FAISS
- [ ] Qdrant
- [ ] Pinecone
- [ ] Weaviate
- [ ] pgvector
- [ ] Not needed

Selected: [Fill here]

Memory : season memory , previous memory , Knowledge memory (RAG) , semantic

4. Design Memory

Memory Needed?

Choose one or more:

- [ ] No memory
- [x] Session memory
- [x] Previous session memory
- [x] Long-term user memory
- [x] Knowledge memory / RAG
- [x] Semantic memory
- [x] Episodic memory
- [x] Procedural memory

Selected: [Fill here]

What Should The Agent Remember?

[Fill here]

What Should The Agent Forget?

[Fill here]

Memory Implementation

Choose one:

- [x] Graph state
- [x] Checkpointer
- [x] LangGraph Store
- [x] Vector database
- [x] App database
- [ ] Not needed

Selected: [Fill here]

5. Create Loop And Patterns

Agent Pattern

Choose one or more:

- [ ] Simple tool-calling agent
- [ ] ReAct
- [ ] Plan and execute
- [ ] Reflection
- [ ] Router

- [x] Human-in-the-loop
- [x] Multi-agent

Selected: [Fill here]

Agent Loop

Write your loop in 4-6 steps:

1. Document agent
2. Fraud agent
3. Credit risk agent
4. Decision agent
5. [Fill here]
6. [Fill here]

Human Approval Needed?

- [x] Yes
- [ ] No

If yes, where? [Fill here]

6. Add Guardrails

Input Guardrails

Choose:

- [x] Validate request
- [x] Block unsafe tasks
- [x] Detect prompt injection
- [x] Ask for missing information

Selected: [Fill here]

Tool Guardrails

Choose:

- [ ] Read-only first
- [ ] Validate tool arguments
- [ ] Restrict tools by role
- [x] Log tool calls

- [x] Limit retries

Selected: [Fill here]

Approval Guardrails

Approval needed before:

- [x] Sending message/email
- [x] Writing to database
- [x] Deleting data
- [ ] Spending money
- [ ] Deploying/changing production
- [ ] Not needed

Selected: [Fill here]

Output Guardrails

Choose:

- [ ] JSON schema
- [ ] Citations
- [x] Confidence score
- [x] Escalate if unsure
- [ ] No unsupported claims

Selected: [Fill here]

7. Test And Iterate

Test Cases

Write 3-5 test cases:

1. Clean Approval
2. Clear Rejection (Low Credit Score)
3. Clear Rejection (Affordability Failure)
4. Manual Review (Borderline Case) - Human in loop
5. Policy Conflict / Priority Logic - Human in loop

Evals Needed

Choose:

- [ ] Programmatic evals
- [x] RAG evals
- [x] Agent/tool evals
- [x] Human review
- [ ] Not for first demo

Selected: [Fill here]

Eval Tool

Choose:

- [ ] LangSmith
- [ ] OpenAI Evals
- [x] Ragas
- [x] pytest
- [x] Custom checklist

Selected: [Fill here]

What Will You Improve After Testing?

[Fill here]

8. Deploy And Monitor

Deployment Type

Choose one:

- [ ] Local demo
- [x] Streamlit / Gradio
- [x] FastAPI
- [ ] Vercel / Render
- [ ] AWS
- [ ] Azure
- [x] Google Cloud
- [ ] Kubernetes
- [ ] Not deploying today

Selected: [Fill here]

Persistence

Choose:

- [ ] No persistence
- [ ] InMemorySaver
- [ ] Postgres checkpointer
- [x] LangGraph Store
- [x] App database
- [x] Vector database

Selected: [Fill here]

Monitoring

Choose:

- [x] Logs
- [x] Traces
- [ ] Tool calls
- [ ] Latency
- [ ] Cost
- [ ] Eval scores

Selected: [Fill here]

Final Demo Plan

Demo Story

In 3-5 lines, explain what you will show.

[Fill here]

Architecture Summary

Model: [Fill here]

Framework: [Fill here]

Interface: [Fill here]

Tools: [Fill here]

Memory: [Fill here]

Guardrails: [Fill here]

Deployment: [Fill here]

Biggest Risk

[Fill here]

What You Completed

[Fill here]

What Is Pending

[Fill here]